

Subject Name: Genetics and Genetic Engineering

Subject No: BT31201

Prerequisites: Biochemistry (BT20203), Microbiology (BT20205)

Objective:

The subject Genetics is the Science that deals with heredity and variation in living organisms. Transmission of traits or characteristic features from parents to offspring or from one generation to the next is known as heredity; while the differences among individual members of a species in population is called variation. Genetics also involves the scientific study of genes or biological information. Genetics is the core of life science, and genes are fundamental to biological information. Understanding the subject genetics and devising genetic tools permit the dissection of biological complexity. Genetics could be regarded as a multidisciplinary subject, as knowledge about other subjects, such as, zoology, botany, microbiology, biochemistry, mathematics and statistics are important to study genetics. We want to utilize these knowledge banks for commercial applications and utilization for our sustenance. In this course, the students will learn the fundamental aspects of genetics, and eventually will be competent to address the biological problems at the molecular level. Genetic engineering deals with the process of gene manipulation combining at least two DNA fragments from two different sources. The process involves isolation of particular DNA fragments (genes), creation of more copies and insertion into other organism via appropriate vectors to obtain enhanced and desired characteristics in living organisms or as their products.

LTP- 3-1-0

Credit- 4

Content:

1. Introduction to the subject “Genetics”; Relationship between particulate factor, gene, biological information, chromosome and cell; Why we want to study Genetics? Different sub-disciplines of Genetics.

[1 lecture]

2. Definitions and concepts of some terms, such as Trait, Genotype, Phenotype, True-breeding or pure line, Crossing or Hybridization, Locus, Allele, DNA polymorphism & its practical utilities etc

[1 lecture]

3. Inheritance pattern of single trait/gene is governed by Mendel’s 1st Law or Principle of Segregation, as established by monohybrid cross.

[1 lecture]

4. Rules of probability in inheritance of traits or genetic analyses; Chi-square (χ^2) test: statistical analyses of the Mendelian monohybrid and dihybrid ratios

[1 lecture]

5. Chromosomal theory of inheritance: Unique features & significance of mitosis and meiosis; How to explain Mendel’s Laws of Segregation and Independent Assortment of alleles of genes from chromosome behavior at meiosis; Sex determination by different systems

[2 lectures]

6. Extension and deviations from Mendelian Inheritance pattern: (A) Interactions between the alleles of a single gene>Extension to Mendel's Law of segregation (B) Interactions amongst the alleles of two genes> Extension to Mendelian law for two-gene inheritance [3 lectures]
7. Linkage, Crossing over & Recombination; Construction of Linkage Map; Chi-square (χ^2) test to determine the linkage between the genes [2 lectures]
8. Chromosomal Variation/Mutation- Numerical and Structural; Transpositions [2 lectures]
9. Population Genetics, Allele frequency and genotype frequency, Hardy-Weinberg equilibrium, Linkage disequilibrium; Mutations, Natural Selection, Genetic drift. [3 lectures]
10. Genetic disorders: Genetic disorders and their genetic basis, Inheritance of genetic diseases. Gene therapy [2 lectures]
11. DNA damage and mutation: Types of mutation, Molecular basis of mutation, mechanism of mutations (spontaneous and induced), DNA damage repair [2 lectures]

Genetic engineering

12. Restriction-modification system, Restriction endonucleases, Ligation, Cloning techniques, Polymerase chain reaction [3 Lectures]
13. Cloning and expression vectors, Prokaryotic and eukaryotic expression systems, Selectable markers and reporter systems [4 lectures]
14. Gene delivery systems, transformation, transfection, viral gene delivery [3 lectures]
15. Common techniques in recombinant DNA technology: DNA isolation, agarose gel electrophoresis, Southern and Northern hybridization, DNA sequencing [4 lectures]
16. Gene silencing and knock out [2 lectures]

Tutorial/Assignments/Exercise/Interactive doubt-clearing sessions – 12 hours

Text books:

1. *Introduction to Genetic Analysis* by Griffiths et al. (W. H. Freeman and Company)
2. *iGenetics: A Molecular Approach* by P. J. Russell (Pearson Education Inc./ Benjamin Cummings)
3. *Genetics: From Genes To Genome* by Hartwell et al. (McGraw-Hill)
4. *Principles of Population Genetics* by Daniel L. Hartl and Andrew G. Clark (Sinauer Associates).
5. *Principles of Gene Manipulation and Genomics* (7th edition): Primose and Tymann, Wiley (2006).
6. *Gene cloning and DNA analysis* (8th edition): Brown, Wiley Blackwell (2020).